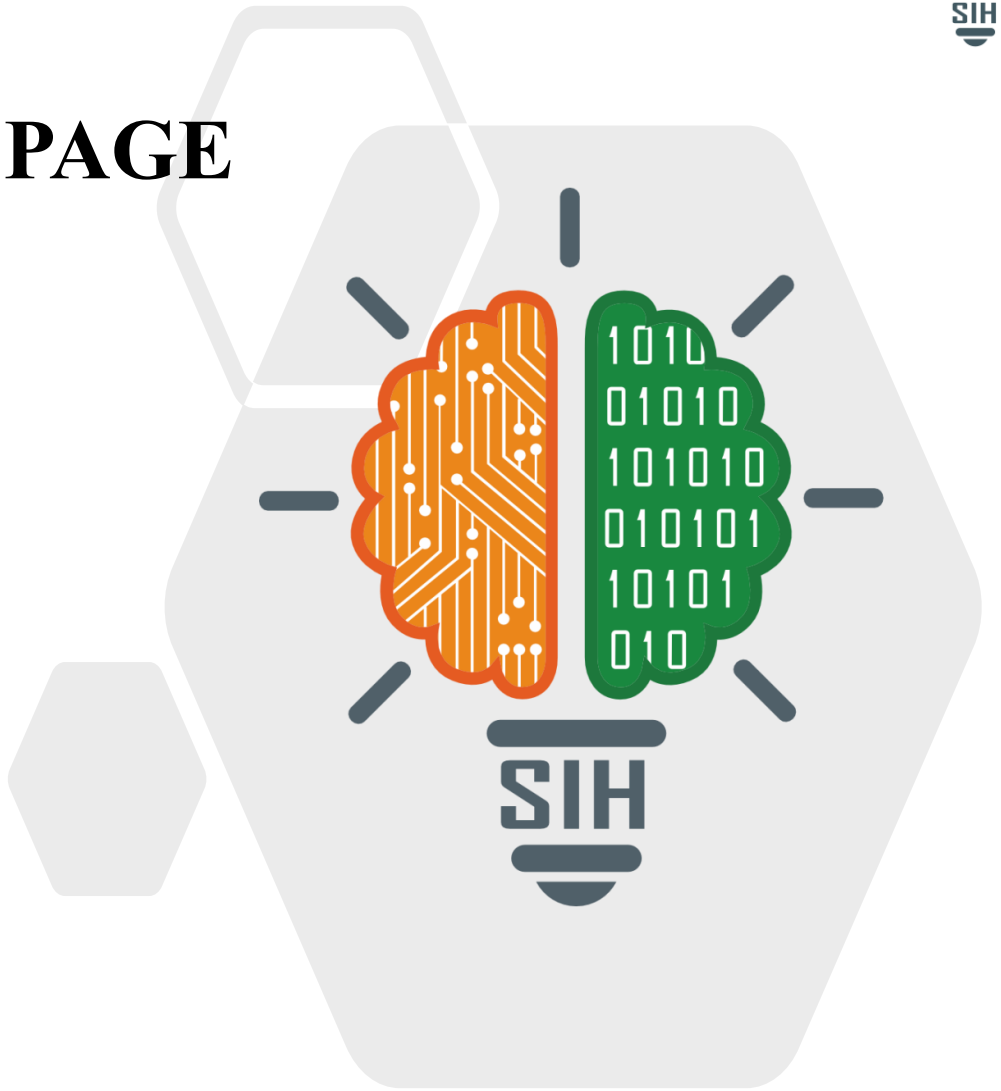
TITLE PAGE

- **Problem Statement ID –** 26172
- **Problem Statement Title –** Low Latency and Efficient Voice Activator for Edge Devices
- **Theme –** Smart Automation
- **PS Category –** Hardware
- **Team ID –** [Team ID]
- **Team Name –** [Team Name]



[Team Name]

Hey Vayro: Edge Wake Word, Cloud ASR

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❖ Proposed Solution (Describe your Idea/Solution/Prototype)

Detailed explanation of the proposed solution

- ESP32-S3 + INMP441 microphone listen continuously; every 10 ms the chip computes a 40-band log-mel frame
- A 10.7 k-parameter DS-CNN (28 KB int8, TensorFlow Lite Micro + ESP-NN) scores the last 1 s every 60 ms
- “Hey Vayro” detected (mean of 3 scores ≥ 0.70) → 571 ms pre-roll + live audio stream over one warm TCP link
- Server: VAD ends the utterance, Vosk (Kaldi) ASR returns the command text

How it addresses the problem

- No audio leaves the device until the keyword is detected → privacy, zero cloud cost while idle
- Inside the PS budget on real hardware: 245.6 KiB peak RAM, 4.7 % idle CPU
- Low latency: first audio at the server ≈ 2 ms after detection; keyword end → server ≈ 95 ms
- Only open-source components; custom keyword trained from scratch (no pre-trained wake word)

Innovation and uniqueness of the solution

- Training and device share one bit-exact feature pipeline (12 000 / 12 000 frames identical)
- Pre-roll ring buffer: a command spoken with no pause after the keyword is never clipped
- Learns from real homes: its own logged false triggers become hard negatives (model v4)
- Each model is chosen by a rule written before testing; 58 logged evidence entries

28 KB	245.6 KiB	4.7 %	≈ 95 ms
int8 model · 13.4 KB tensor arena	peak RAM while streaming (< 256 KB)	idle CPU, both cores (busiest core 8.8 %)	keyword end → server (median)

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TECHNICAL APPROACH

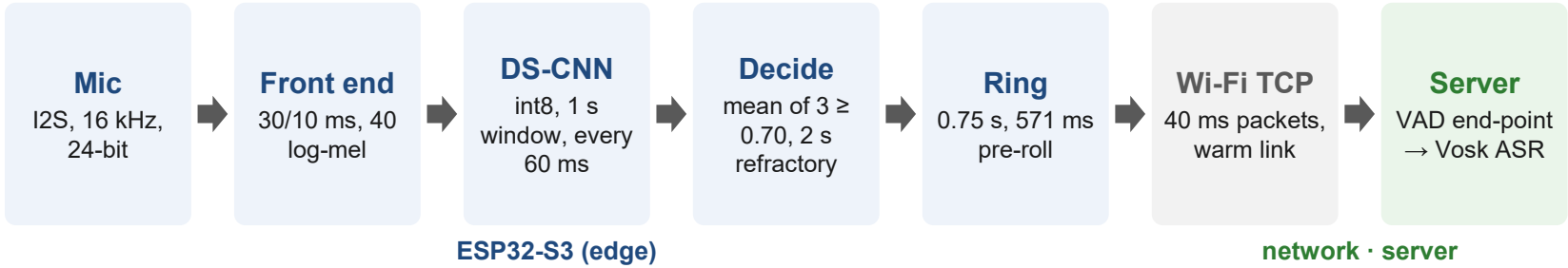
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❖ **Technologies to be used (e.g. programming languages, frameworks, hardware)**

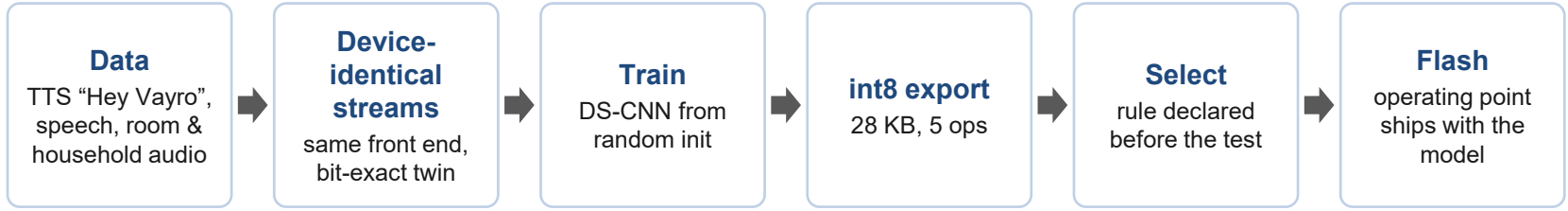
- **Hardware:** ESP32-S3 N16R8 (240 MHz, PSRAM off) + INMP441 I2S MEMS mic
- **Firmware:** C / C++ on ESP-IDF 5.5.5 (FreeRTOS)
- **TinyML:** TensorFlow Lite Micro + ESP-NN kernels, TFLM microfrontend
- **Training:** Python, TensorFlow / Keras, Piper TTS (4 voices), LibriSpeech, our own recordings
- **Server:** Python TCP ingest, energy VAD, Vosk / Kaldi ASR
- **Verification:** bit-exact host twin of the front end, scripted capture and replay tools

❖ **Methodology and process for implementation (Flow Charts/Images/ working prototype)**

Runtime — on the ESP32-S3 until the keyword, then the server



How the keyword model is made (all scripted, reproducible)



Working prototype: ESP32-S3-N16R8 + INMP441 on a breadboard (SCK→GPIO6, WS→GPIO5, SD→GPIO4). The demo video shows keyword → stream → transcript live on this board.

Model: 10 665 parameters · 212 584 MACs per inference · 28 024 B int8 · tensor arena 13 404 B · ops: Conv2D, DepthwiseConv2D, FullyConnected, Reshape, Logistic

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FEASIBILITY AND VIABILITY

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❖ Analysis of the feasibility of the idea — measured on the physical prototype, final firmware

Efficiency

245.6 KiB peak RAM, streaming (< 256 KB)	**4.7 %** idle CPU, 2 cores (max core 8.8 %)	**28 KB** model flash · 13.4 KB arena

Accuracy (model v4)

21 / 25 held-out keyword prompts, 0.5–2 m	**10 / 10** fresh keyword + command prompts	**0 in 5 h** false triggers, quiet night (held-out)

Latency

1.7 ms detection → first packet at server	**≈ 95 ms** keyword end → server (median)	**136 ms** keyword end → server (p95)

❖ Potential challenges and risks	❖ Strategies for overcoming these challenges
TV / family speech still triggers: 10.4 per hour (held-out household audio)	Hard negatives mined from the device's own false triggers already cut 15.1 → 10.4 /h; next: more household hours, server-side second check that drops streams with no command
Real-voice testing so far: one main speaker (plus TTS voices in 4 accents)	Scripted 3-minute sessions for new speakers; per-speaker results reported separately
Server ASR word accuracy is set by the ASR model (Vosk en-in: 36 % WER published)	Edge is ASR-agnostic: swap in a larger model or a command grammar on the server only
Thin RAM margin; Wi-Fi or server outages	Memory trims saved 9.6 KB; detection keeps running offline; auto-reconnect; ASR off the server's receive path

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IMPACT AND BENEFITS

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❖ Potential impact on the target audience

- **ISRO labs & control rooms:** hands-free voice commands where hands and eyes are busy — without streaming room audio all day
- **Smart automation:** a ready wake-word front end for any ESP32-class home, office or factory device
- **Assistive use:** voice control for elderly or motor-impaired users on low-cost hardware
- **Indian developers & start-ups:** custom keyword with Indian-accent training data, no vendor SDK or licence fee

0 bytes
of audio sent until the keyword is detected

32 kB/s
16 kHz audio, only from keyword to end of command

❖ Benefits of the solution (social, economic, environmental, etc.)

Social

Privacy by design: the room is only heard by the chip until someone says “Hey Vayro”

Economic

Hardware ≈ ₹1 000 (estimate); open-source stack; no cloud cost while listening

Environmental

Low compute (4.7 % CPU idle); audio is sent only after the keyword, so traffic stays tiny. Wi-Fi is kept awake for latency; power not yet measured

Technical

Reproducible: a new keyword = new recordings + the same scripts; every number traceable to an evidence log

[Team Name]

RESEARCH AND REFERENCES

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❖ Details / Links of the reference and research work

[1] **Y. Zhang, N. Suda, L. Lai, V. Chandra,** “Hello Edge: Keyword Spotting on Microcontrollers”, arXiv:1711.07128, 2017 — DS-CNN architecture

[2] **P. Warden,** “Speech Commands: A Dataset for Limited-Vocabulary Speech Recognition”, arXiv:1804.03209, 2018 — KWS background (not used for training)

[3] **TensorFlow Lite Micro + microfrontend:** github.com/tensorflow/tflite-micro

[4] **Espressif esp-tflite-micro and ESP-NN:** github.com/espressif/esp-tflite-micro · github.com/espressif/esp-nn

[5] **ESP-IDF v5.5 (FreeRTOS, drivers, lwIP):** github.com/espressif/esp-idf

[6] **V. Panayotov et al.,** “LibriSpeech: an ASR corpus based on public domain audio books”, ICASSP 2015 — openslr.org/12

[7] **Piper TTS and voices:** github.com/OHF-Voice/piper1-gpl · huggingface.co/rhasspy/piper-voices

[8] **D. Povey et al.,** “The Kaldi Speech Recognition Toolkit”, ASRU 2011 · Vosk: alphacephei.com/vosk

[9] **Datasheets:** ESP32-S3 Technical Reference Manual (Espressif); INMP441 (TDK InvenSense)

Our own research record

- **Evidence ledger:** 58 entries, each tagged measured / estimated / proposed, with its raw data
- **Selection rules:** written and committed before each held-out test
- **SBOM:** every component and licence listed; the one closed Wi-Fi radio blob and the non-commercial TTS voices are disclosed
- **Licence:** our code Apache-2.0; every component’s licence listed in the SBOM
- **Demo video:** link to be added